

UNDERSTANDING Mathematical Induction

Prove It for All n !

Grade 6 Mathematics | Classroom Lesson

"Induction is a magical ladder to climb infinitely many steps!"

1. What Does "For All n " Mean?

"For all n " means the statement is true for **every natural number** starting from 1, 2, 3, and continuing forever — no matter how large n gets.

KEY IDEA

We do not test just one or two numbers.

We prove it works for 1, 2, 3, 4, 5 ... and every number after that.

It includes every natural number, no matter how big!

Think of it like this:

Imagine you are jumping up a staircase. If you can jump 1 step, then 2 steps, then 3 steps ... can you jump ALL the steps? Mathematical induction says YES — and gives you a rigorous way to prove it.

2. Why Do We Need to Prove for All n ?

Sometimes we want to show things are true **forever**, not just for one number. Mathematical induction helps us prove that patterns work for every single natural number.

IT HELPS US...

- Understand patterns in mathematics
- Prove infinite families of facts
- Build strong reasoning skills
- Verify formulas we discover

WITHOUT INDUCTION...

- We could only check finitely many cases
- We could never be 100% sure for ALL n
- Patterns might break at some large n
- Proofs would be incomplete

3. What Is Mathematical Induction?

Mathematical induction is a **two-step proof technique** used to prove statements about all natural numbers.

Mathematical induction is like climbing stairs: if you can reach the first step, and from any step you can reach the next — you can climb ALL stairs!

1

Base Case (First Step)

Prove the statement is true for $n = 1$.

2

Inductive Step (Next Step)

Assume true for $n = k$, then prove true for $n = k + 1$.

RESULT

Together, these two steps prove the statement is true for ALL $n \geq 1$.

4. Step 1 — The Base Case

First, we prove the statement is true for the **smallest value**, usually $n = 1$. This is like making sure you can step onto the first stair before climbing higher!

HOW TO DO THE BASE CASE

Substitute $n = 1$ into the left-hand side (LHS).

Substitute $n = 1$ into the right-hand side (RHS).

Show $LHS = RHS$. If they match, the base case is proven!

Example: Prove that $1 = 1(1+1)/2 \Rightarrow LHS = 1, RHS = 1(2)/2 = 1$. Since $1 = 1$, the base case holds. ✓

Remember: The base case is your *starting point*. Without it, the induction chain has no foundation!

5. Step 2 — The Inductive Step

Next, we assume the statement is true for $n = k$ (called the *Inductive Hypothesis*), and then prove it must also be true for $n = k + 1$. This is the magic step that connects everything together!

INDUCTIVE HYPOTHESIS

Assume the statement holds for $n = k$.

Write it down clearly.

This is NOT a proof — it is an assumption.

We will USE this assumption next.

PROVE FOR $K + 1$

Start with the expression for $n = k+1$.

Use the inductive hypothesis to simplify.

Algebra should give the RHS for $n = k+1$.

Conclude the step is proven.

6. Putting It Together — Full Proof Template

Follow these four steps for any induction proof:

1

Prove the Base Case ($n = 1$)

Show LHS = RHS when you substitute $n = 1$.

2

State the Inductive Hypothesis ($n = k$)

Assume the statement is true for $n = k$.

3

Prove for $n = k + 1$

Using the hypothesis, prove the statement holds for $k + 1$.

4

Write the Conclusion

By induction, the statement is true for all $n \geq 1$.

7. Example 1 — Sum of First n Natural Numbers

Statement to prove:

$$1 + 2 + 3 + \dots + n = n(n + 1) / 2 \text{ for all } n \geq 1$$

STEP 1: BASE CASE (N = 1)

$$\text{LHS} = 1$$

$$\text{RHS} = 1(1 + 1) / 2 = 1(2) / 2 = 1$$

Since LHS = RHS = 1, the base case holds. ✓

STEP 2: INDUCTIVE HYPOTHESIS

Assume the formula is true for $n = k$:

$$1 + 2 + 3 + \dots + k = k(k + 1) / 2$$

STEP 3: PROVE FOR $N = K + 1$

We need to show: $1 + 2 + \dots + k + (k+1) = (k+1)(k+2) / 2$

Starting from LHS:

$$= [1 + 2 + \dots + k] + (k+1)$$

$$= k(k+1)/2 + (k+1) \quad [\text{using the inductive hypothesis}]$$

$$= (k+1) [k/2 + 1]$$

$$= (k+1)(k+2) / 2 \quad [\text{this is RHS for } n = k+1] \quad \checkmark$$

CONCLUSION

By mathematical induction, $1 + 2 + 3 + \dots + n = n(n+1)/2$ for all $n \geq 1$.

8. Example 2 — Multiples of 3

Statement to prove:

$$3n \text{ is a multiple of 3 for all } n \geq 1$$

STEP 1: BASE CASE (N = 1)

$$3 \times 1 = 3$$

$$3 / 3 = 1 \text{ (no remainder)}$$

So 3 is a multiple of 3. ✓

Base case is TRUE!

STEP 3: INDUCTIVE STEP $K \rightarrow K+1$

Assume $3k$ is a multiple of 3.

$$3(k+1) = 3k + 3$$

$3k$ is a multiple of 3 (hypothesis).

3 is a multiple of 3. Sum is also. ✓

CONCLUSION

By induction, $3n$ is a multiple of 3 for all $n \geq 1$.

9. Example 3 — Powers of 2

Statement to prove:

$$2^n \geq n + 1 \text{ for all } n \geq 1$$

STEP 1: BASE CASE (N = 1)

$$\text{LHS} = 2^1 = 2$$

$$\text{RHS} = 1 + 1 = 2$$

Since $2 \geq 2$, base case holds. ✓

STEP 3: INDUCTIVE STEP $K \rightarrow K+1$

Assume $2^k \geq k + 1$.

$$2^{(k+1)} = 2 \times 2^k \geq 2(k+1)$$

$$2(k+1) = 2k + 2 \geq k + 2$$

$$\text{So } 2^{(k+1)} \geq (k+1) + 1 \text{ ✓}$$

CONCLUSION

By induction, $2^n \geq n + 1$ for all $n \geq 1$.

10. Common Mistakes to Avoid

Mistake	Explanation
■ Forgetting the Base Case	You MUST prove $n = 1$ works. Without this, the chain has no starting point.
■ Not Stating the Hypothesis Clearly	Write the inductive hypothesis explicitly before using it.
■ Skipping from k to $k+1$	Show every algebra step. Jumping steps loses marks and clarity.
■ Assuming What You Need to Prove	Do not write "assume the result holds for $n = k+1$ " — that is circular!
■ Forgetting the Conclusion	Always finish with: "By induction, the statement holds for all $n \geq 1$."

11. Try It Yourself — Practice Challenges!

CHALLENGE 1: SUM OF ODD NUMBERS

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$

Hint: Check $n = 1$ first, then assume true for k and prove for $k + 1$!

CHALLENGE 2: POWERS OF 5

5^n is divisible by 5 for all $n \geq 1$

Hint: What is 5^1 ? Show $5^{k+1} = 5 \times 5^k$.

CHALLENGE 3: SUM OF FIRST n SQUARES

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Hint: Add $(k+1)^2$ to both sides of the hypothesis!

12. Recap — What We Learned Today

Concept	Summary
"For all n "	The statement must hold for 1, 2, 3, 4 ... every natural number.
Base Case	Always start by proving the statement is true for $n = 1$.
Inductive Hypothesis	Assume the statement is true for some $n = k$.
Inductive Step	Use the hypothesis to show the statement holds for $n = k + 1$.
Conclusion	Base case + Inductive step $\Rightarrow$ True for ALL $n \geq 1$!

13. Why Does Induction Work?

Induction works because once we prove the first step is true, and we show that **any step leads to the next**, we create an **unbreakable chain of truth** that goes on forever!

Think of falling dominoes: if the 1st falls, and each domino knocks the next — ALL dominoes fall!

You can now climb the **infinite staircase** of mathematics!
Keep practising and you will master proving things "for all n " in no time!
